Microbiology: Bacterial Respiratory Pathogens II

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Session Objectives

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1. List and differentiate bacterial infections of the respiratory system and explain the diseases they cause.
2. Describe the unique microbiologic features and any associated virulence factors for each pathogen
3. Explain any relevant symptoms and diagnostic tests for each pathogen.
4. Describe any treatment or prevention strategy for each pathogen.

Outline

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- Healthcare-Associated and Ventilator Associated Pneumonia
  - *Pseudomonas aeruginosa (also structural lung disease with CF)*
  - *Acinetobacter baumannii*
  - *Escherichia coli (and other enteric bacteria)*
  - *Serratia marcescens*
  - *Methicillin-resistant staphylococcus aureus*
- Aspiration Pneumonia
  - *Actinomyces israelli*
- Immunocompromised Hosts/Structural Lung Disease
  - *Nocardia asteroides*
  - *Mycobacterium tuberculosis*
  - *Burkholderia cepacia (cystic fibrosis)*
- Zoonotic/Occupational Exposures
  - *Coxiella burnetii*
  - *Bacillus anthracis*
  - *Francisella tularensis*
  - *Yersinia pestis*

Healthcare-Associated and
Ventilator Associated Pneumonia

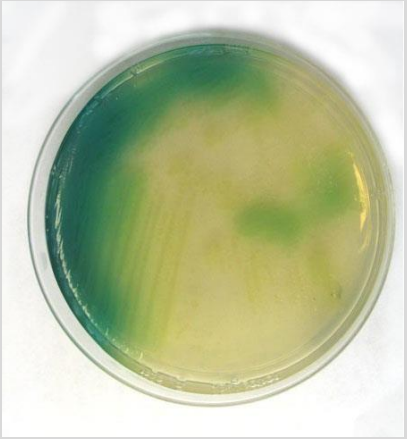
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- ***Pseudomonas aeruginosa***
- ***Acinetobacter baumannii***
- ***Escherichia coli (and other enteric bacteria)***
- ***Serratia marcescens***
- ***Methicillin-resistant staphylococcus aureus***

Pseudomonas aeruginosa

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- **Characteristics:**
  - Gram –ve bacillus, aerobic, motile
  - Oxidase & Catalase +ve
  - Blue-green pigment (pyocyanin)
  - Grape-like odor
  - Likes moist environments
- Virulence
  - Encapsulated
  - LPS Endotoxin (fever, shock)
  - Exotoxin A (inactivates EF-2)
- Associated with Cystic Fibrosis patients



PSEUDOMONAS is associated with:
Pneumonia
Sepsis
Ecthyma gangrenosum
UTI
Diabetes
Osteomyelitis
Mucoid polysaccharide capsule
Otitis externa
Nosocomial infx
Addiction (IV drug use)
Skin infx (hot tub folliculitis)

Acinetobacter baumannii

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- Characteristics:
  - Gram –ve coccobacillus
  - Aerobic
  - Catalase +ve
  - Associated with Healthcare-associated infection via respiratory equipment and indwelling catheters
- High resistance

Escherichia coli

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- Characteristics:
  - Gram –ve bacillus
  - Pathogen within and outside of the GI tract

- Virulence:
  - Fimbriae (P pili) → cystitis and pyelonephritis
  - K-capsule → pneumonia, neonatal meningitis
  - LPS endotoxin → septic shock

Serratia marcescens

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- Characteristics:
  - Gram –ve bacillus
  - Slow lactose fermenter
  - Catalase +ve
  - Red pigment
- Drug resistant
- Most commonly causes UTI, but also pneumonia

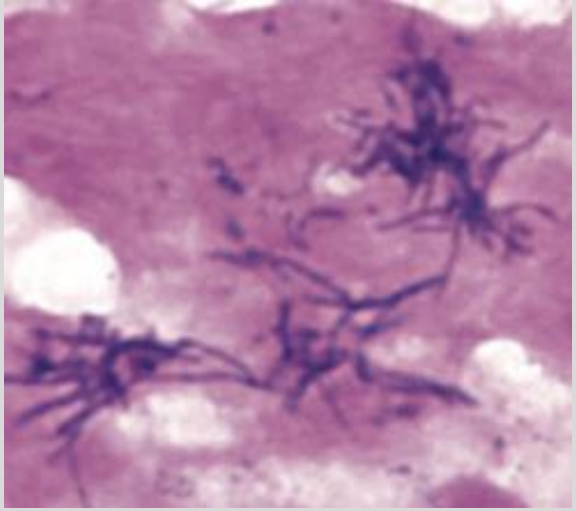


Methicillin-resistant Staphylococcus aureus

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- Review Characteristics:
  - Gram +ve, beta-hemolytic, catalase +ve, coagulase +ve cocci in clusters
- Important cause of serious health-care associated infx
- Resistance: altered penicillin-binding proteins due to *mecA* gene
- Panton-Valentine leucocidin (PVL): seen in some strains which kills leukocytes → tissue necrosis

Actinomyces israelli



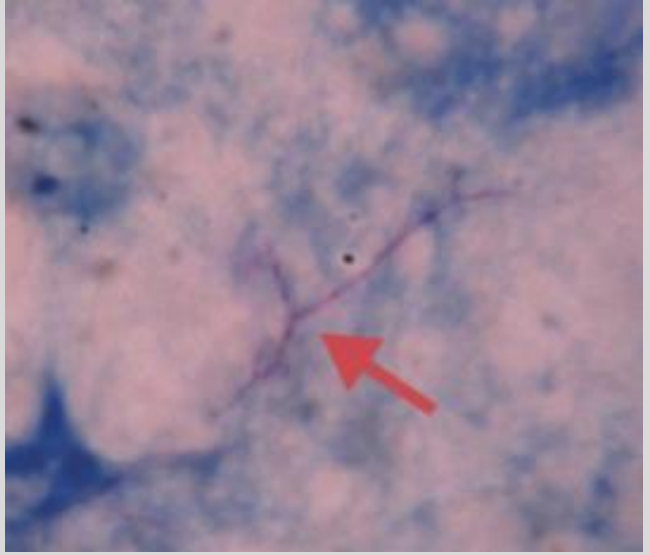
- **Aspiration pneumonia**
- Characteristics
  - Gram +ve Branching filaments
  - Anaerobe
  - Catalase -ve
  - Produces yellow “sulfur” granules which are made of filaments of bacteria
  - Normally in GI tract/mouth and usually associated with dental caries/extractions; if aspirated→ pneumonia
- Tx: Penicillin

Immunocompromised Host/Structural Lung Disease

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- ***Nocardia asteroides***
- ***Mycobacterium tuberculosis***
- ***Burkholderia cepacia*** *(cystic fibrosis)*

Nocardia asteroides

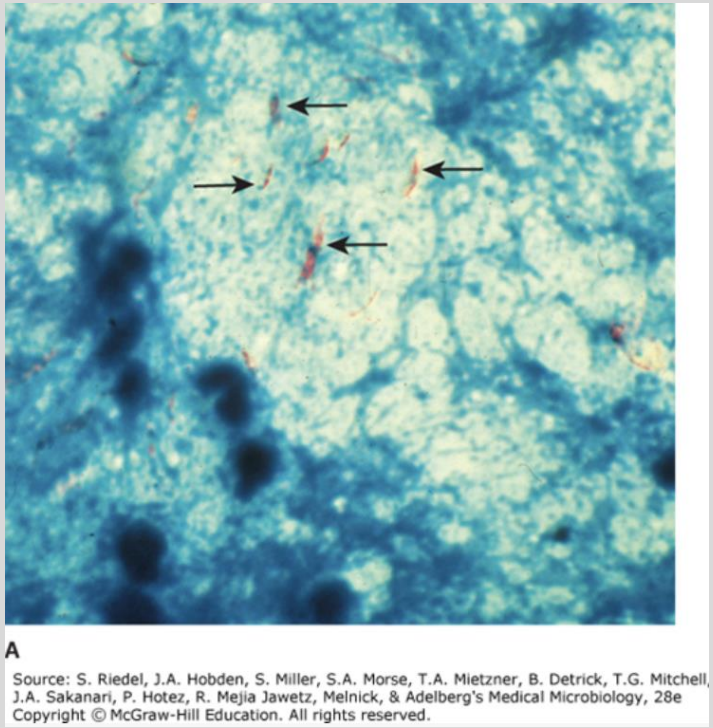


- Characteristics:
  - Gram +ve; Branching filament
  - Aerobic; non-motile
  - Catalase +ve
  - Weakly acid fast (mycolic acid in cell wall)
  - Found in soil; transmitted via inhalation
- Virulence:
  - "**cord factor**" (CF): facilitates intracellular survival
- Causes pulmonary infections in **immunocompromised** that can disseminate to other organs (ex: brain, skin)
- Tx: sulfonamides

Mycobacterium tuberculosis

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- Characteristics:
  - Rod shaped, aerobic
  - Does not Gram stain well
  - Auramine-Rhodamine stain can replace acid-fast
  - Cultures on Lowenstein-Jensen medium
  - Facultative intracellular
- Virulence:
  - cord factor: activates macrophages (promoting granuloma formation) and induces release of TNF-α
  - Sulfatides: inhibit phagolysosomal fusion



A

Source: S. Riedel, J.A. Hobden, S. Miller, S.A. Morse, T.A. Mietzner, B. Detrick, T.G. Mitchell, J.A. Sakanari, P. Hotez, R. Mejia Jawetz, Melnick, & Adelberg's Medical Microbiology, 28e
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Mycobacterium tuberculosis

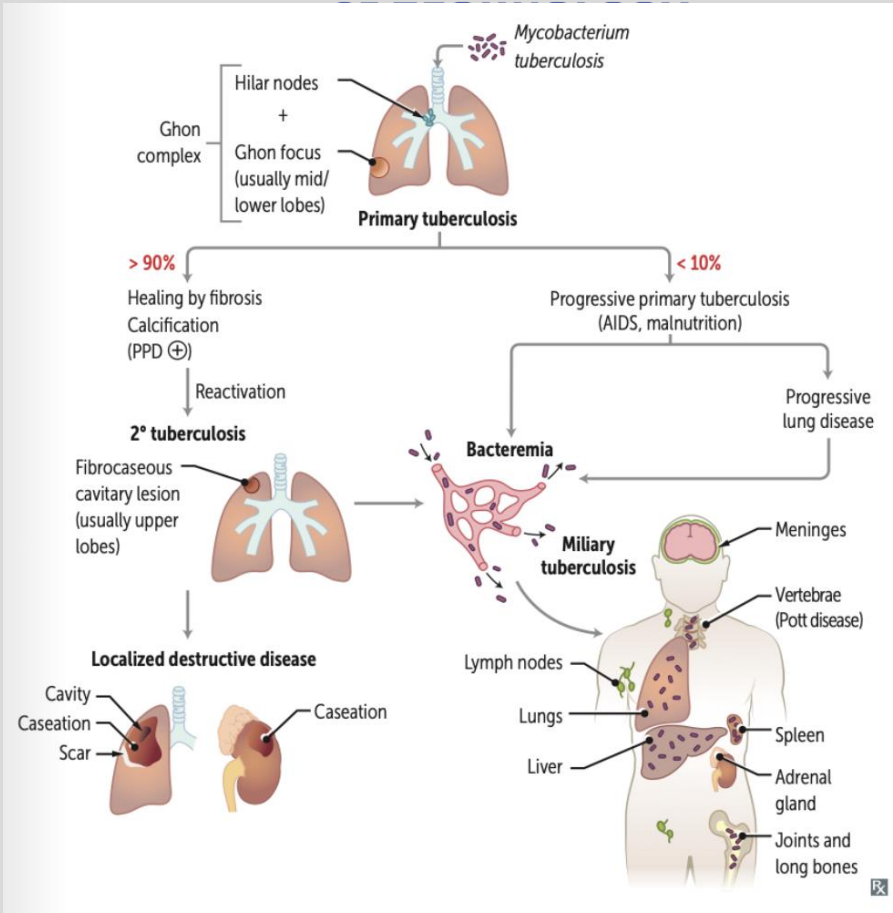
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- Tuberculosis:
  - Transmission: inhalation of infected droplet
  - High Risk Exposure: Prisoner, homeless, healthcare workers, immigrants, IVDU
  - High Risk for Progression: HIV, history of prior or untreated TB, IVDU, age extremes
  - s/sx:
    - Most initial active TB, and latent TB, is asymptomatic
    - If primary progressive or if reactivated TB: fever, night sweats, weight loss, cough, hemoptysis

Mycobacterium tuberculosis

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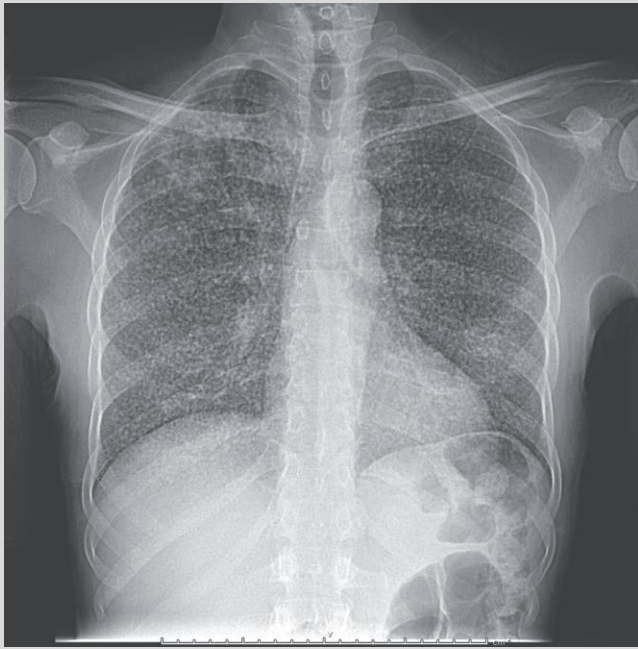
- Tuberculosis Pathogenesis
  - Inhaled mycobacteria → engulfed by alveolar macrophages and replicate
    - More lymphocyte and macrophage recruitment → CASEATING GRANULOMA
    - GHON FOCUS + HILAR LYMPHADENOPATHY = GHON COMPLEX (primary TB)
  - About 90% of people with Primary TB → <u>Latent TB</u> (Dormant; not infectious)
  - Less than 10% may progress → <u>Primary progressive TB</u>
  - If spreads to bloodstream → <u>Miliary or Extrapulmonary TB</u>



Mycobacterium tuberculosis



- Tuberculosis Diagnosis
  - Tuberculin Skin Test
    - Measure induration 48-72 hours after injection
    - + PPD: active or latent TB; BCG vaccine (false positive)
  - Interferon Gamma Release Assay
    - Good for those who are likely to have false PPD results
  - Sputum sample and culture
  - imaging with chest X-ray
    - If negative → latent TB
    - If necrosis of upper lobes → Active TB (primary progressive or reactivated TB)
    - If “millet seeds” diffusely across lungs → Miliary TB



Mycobacterium tuberculosis

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- Tuberculosis Treatment
  - Tx: Latent TB:
    - **I**soniazid for 9 months OR
    - **R**ifampin/rifamycin for 4 months
  - Tx: Active TB: **RIPE** for 2 months and RI for 4 months
    - **R**ifampin
    - **I**soniazid
    - **P**yrazinamide
    - **E**thambutol

Burkholderia cepacia

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- Characteristics:
  - Gram –ve bacillus
  - Aerobic
  - Catalase +ve
  - Oxidase +ve
- Causes pneumonia in pts with underlying lung disease, such as Cystic Fibrosis
- Often multidrug resistant

Zoonotic/Occupational Exposures

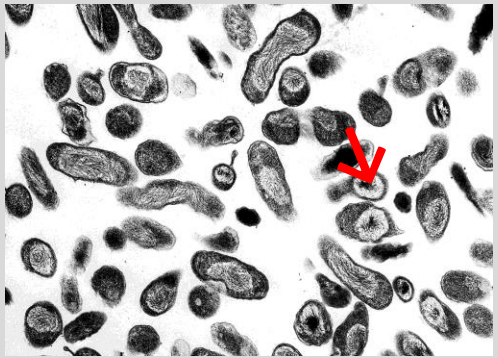
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- ***Coxiella burnetii***
- ***Bacillus anthracis***
- ***Francisella tularensis***
- ***Yersinia pestis***

Coxiella burnetii

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- Characteristics:
  - Considered gram –ve but doesn't Gram stain well
  - Obligate intracellular: grows/replicates inside host cells
  - "*Rickettsia*-like"; No vector
  - Is a bacteria but has atypical spore-like structure

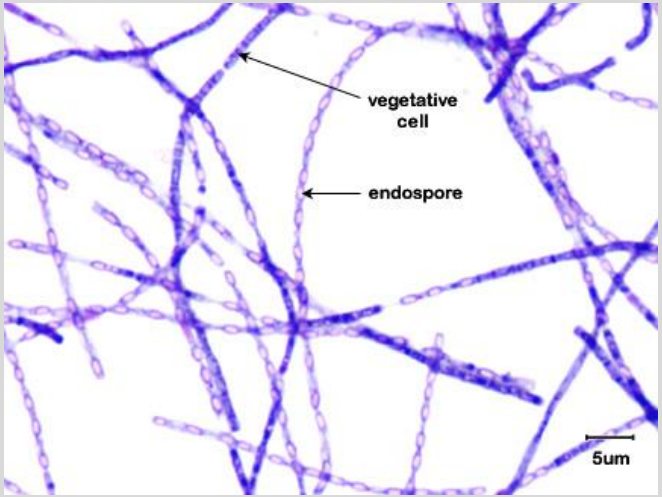


- <u>Q-fever:</u>
  - Inhaled as aerosol from cattle/sheep amniotic fluid/urin
  - S/Sx: headache, cough, flulike sx, atypical pneumonia
  - May lead to endocarditis, pneumonitis, and hepatitis
  - Tx: doxycycline

Bacillus anthracis

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- Characteristics:
  - Gram +ve
  - Endospore forming (need to autoclave to kill)
  - Obligate aerobe
  - Poly D-glutamate capsule



- Anthrax Toxin:
  - Protective antigen: forms membrane channels for lethal and edema factors to enter cell
  - Lethal toxin: inhibits MAP kinase → inhibits growth, leads to apoptosis
  - Edema toxin: increases cAMP → tissue edema

Bacillus anthracis

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- Anthrax:
  - Infection through spore inhalation, ingested, or injured skin from infected animals
  - Cutaneous Anthrax: painless, necrotic papule surrounded by vesicles→ black eschar
  - **Pulmonary Anthrax:** inhalation of spores (from animals; also, a bioweapon) → flulike sx that rapidly progress to fever, pulmonary hemorrhage, mediastinitis and shock
    - Rapidly fatal
    - AKA Wool-sorter's disease
    - Tx: ciprofloxacin



Francisella tularensis

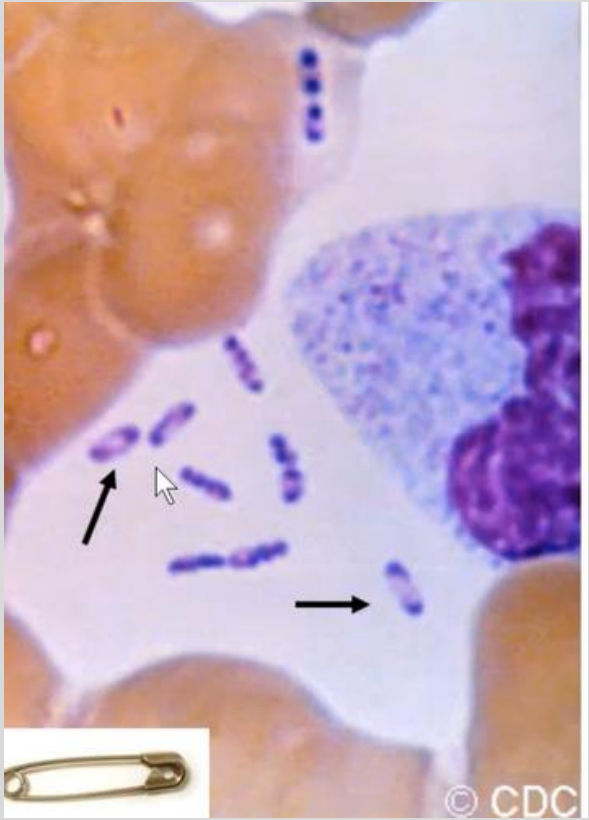
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- Characteristics:
  - Gram –ve Coccobacillus
  - Facultative intracellular
  - Two biotypes
- Tularemia:
  - Transmitted through ticks, rabbits, deer flies
  - After bite, regional lymph nodes enlarge → necrotic and painful; fever, malaise, headache
  - Inhalation of infective aerosol → pneumonia
  - Tx: streptomycin

Yersinia pestis

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- Characteristics:
  - Gram –ve
  - Bacilli
  - Nonmotile
  - Facultative anaerobe
  - Facultative intracellular
  - Safety pin-like appearance
  - Transmitted between rodents & fleas



Yersinia pestis

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- **Bubonic Plague:**
  - 40-100% mortality
  - s/sx:
    - Sudden onset of high fever, chills, weakness, rapid progression to septic shock
    - Intense inflammation develops in enlarged lymph nodes→ buboes
    - Can stay in lymph nodes, but often reaches bloodstream → meningitis, pneumonia, septic shock, or multi-organ failure
  - **Primary pneumonic plague:**
    - inhale infective droplets from coughing patient
  - Tx: streptomycin



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Questions for review- answers in notes section of PPT

Practice Question #1

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- **Q:** An 80-year-old female with PMHx of T2DM and dementia is brought to the hospital due to three weeks of persistent fever, pleuritic chest pain, and cough despite treatment with levofloxacin for suspected pneumonia. Currently her vitals are T:100.8 F, BP 130/78; HR: 88; RR: 18; O2: 96% RA. On physical exam, the patient has decreased breath sounds of the right lower lung field. Chest CT is ordered and reveals right lower lobe infiltrate, right pleural effusion, and right-sided hilar lymphadenopathy. What is the most likely cause of this patient's condition?

  A. Mycobacterium tuberculosis

  B. Streptococcus pneumoniae

  C. Escherichia coli

  D. Staphylococcus aureus

  E. Mycoplasma pneumoniae

Practice Question #2

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- Q: A 44-year-old wool worker develops fever, malaise, and rapidly progressive shortness of breath. Chest X-ray reveals a widened mediastinum. Blood culture grows a large, Gram-positive rod. Which of the following virulence factors explains the pathogenicity of this organism?

  A. Protein A that binds Fc region of IgG
  B. Polysaccharide capsule resisting opsonization
  C. Poly-D-glutamate capsule inhibiting phagocytosis
  D. Lipooligosaccharide endotoxin activating macrophages
  E. Heat-labile toxin increasing cAMP

Practice Question #3

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- Q: A 42-year-old farmer presents to the hospital with two days of fever, severe headache, and cough. The patient states he started feeling symptoms a day after he had to assist with delivering a sheep. Gram stain only shows inflammatory cells. Which of the following organisms is the most likely cause of this patient's illness?

  A. Francisella tularensis
  B. Yersinia pestis
  C. Coxiella burnetii
  D. Bacillus anthracis
  E. Pasteurella multocida

Summary Slide

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- Healthcare-Associated and Ventilator Associated Pneumonia
  - *Pseudomonas aeruginosa:* Gram –ve rod; catalase +ve, oxidase +ve; aerobic; motile; grape odor; moist environments; blue-green pigment on agar; seen in CF pts
  - *Acinetobacter baumannii:* Gram -ve; aerobic; catalase +ve; high resistance
  - *Escherichia coli:* Gram –ve rod; indole +; mannitol fermenter; K-capsule virulence causes pneumonia
  - *Serratia marcescens:* Gram -ve; catalase +ve; slow fermenter on MacConkey; red pigment
  - *Methicillin-resistant staphylococcus aureus:* Gram +ve cocci in clusters; β-hemolytic; catalase +ve; coagulase +ve; resistance through altered PBP's due to *mecA* gene
- Aspiration Pneumonia
  - *Actinomyces israelli:* Gram +ve branching filaments; anaerobe; yellow sulfur granules; associated with dental caries/extractions and if aspirated → pneumonia

Summary Slide

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- Immunocompromised Hosts/Structural Lung Disease
  - *Nocardia asteroides:* Gram +ve branching filament; aerobic; non-motile; catalase +ve; weakly acid fast; cord factor virulence; inhaled soil; pulm infx seen in immunocompromised

  - *Mycobacterium tuberculosis:* aerobic rod; does not Gram stain well; acid fast; Lowenstein-Jensen medium; facultative intracellular in alveolar macrophages; virulence through cord factor and sulfatides; caseating granulomas; ghon focus + hilar adenopathy = ghon complex in primary TB; miliary TB; Tuberculin skin test; IFN gamma; RIPE tx

  - *Burkholderia cepacia:* Gram –ve rod; aerobic; catalase +ve; oxidase +ve; pneumonia in pt with CF

Summary Slide

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- Zoonotic/Occupational Exposures
  - *Coxiella burnetii:* doesn't gram stain well; obligate intracellular; Q-fever
  - *Bacillus anthracis:* Aerobic Gram +ve spore forming; poly D-glutamate capsule; Lethal toxin; edema toxin; Pulmonary anthrax
  - *Francisella tularensis:* Gram –ve coccobacillus; facultative intracellular; ticks and wild rabbits; Tularemia
  - *Yersinia pestis:* Gram –ve bacillus; nonmotile; facultative intracellular anaerobe; safety pin-like appearance; The Plague

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THANK YOU!

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